

DARREN L. OLDAG

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SUMMARY

Software architect with over 10 years experience and a proven track record of reliable and maintainable solutions. Excels at designing robust and extensible systems with a keen understanding of the given problem and an eye for future enhancement. Responsible for a wide variety of projects in multiple areas. Specializes in object-oriented design and development (Java, C++), but possesses experience in many languages (C, Perl, assembly, etc.). Cross-platform developer for assorted operating systems (Linux, Windows, Mac, Solaris) and processor architectures (x86, PowerPC, Sparc).

EXPERIENCE

Oracle ← Sun Microsystems ← MySQL, Inc. , Austin, Texas

Senior Software Developer, May 2006 to Present

- Key contributor to the MySQL Enterprise Tools team responsible for developing MySQL Enterprise Monitor.
- Technical owner of the Java server backend for MySQL Enterprise Monitor.
 - Spearheading the effort to modularize the monolithic web application using an OSGi runtime/container in order to improve manageability and maintainability.
 - Redesigning the rules engine according to flow-based programming (dataflow) principles.
 - Designed and implemented the REST interface to MySQL Enterprise Monitor's backend data model.
 - Architect of the persistence and service layers for reporting and analysis of queries across one or many MySQL database servers.
 - Replaced direct database access with Hibernate object-relational mapping layer.
 - Refactored application to use Spring Framework for dependency injection and inversion of control.
 - Implemented replication topology discovery based on simple master/slave relationships, allowing replication server grouping and visualization.
 - Added multi-variable, expression-based graphing support, enabling pluggable graphing for internal and external customer use.
 - Profiled and optimized the backend through several iterations and releases.
 - Discovered and decreased memory waste, including improving internal Hibernate caching inefficiencies.
 - Identified and reduced thread concurrency and scalability hotspots.
 - Performed SQL query analysis and optimization.
- Developed MySQL Sandbox extension to deploy replication and MySQL proxy topologies specified in the DOT graph description language.

Simdesk, Austin, Texas

Senior Software Engineer, August 2005 to April 2006

- Designed and implemented enhancements to the Simdesk World Wide Server™ Personal Information Management (PIM) component, including email distribution list storage and synchronization, attendee list management for meeting support, and extensible properties for any PIM object.
- Designed a calendaring object model conformant to the RFC 2445 iCalendar specification.

Newisys, a Sanmina-SCI Company, Austin, Texas

Design Verification Engineer, September 2004 to July 2005

- Ownership, including design and implementation (Java), of testbench and testcases for simulation and verification of a hardware (Verilog) memory controller module used in a custom ASIC.

Pervasive Software, Austin, Texas

Senior Software Engineer, March 2000 to August 2004

- Ownership and development of Pervasive Services, a platform abstraction layer providing a common API and class library for development on Linux, Win32, Solaris, Netware, and MacOS X. Library includes functionality for abstracting processor architecture (x86, PowerPC, Sparc), file handling, IPC, process control, threading, synchronization, etc.
- Maintenance and extension of cross platform component build system based on GNU Make. Extensions include customizable frameworks, seamless external and cross-tree consumption, and Java build integration (both native and Ant).
- Designed and implemented a pluggable and skinnable test framework, thus providing a common infrastructure for unit and functional tests. Common framework allowed for automatable, repeatable results, reducing the necessity for quality assurance and developer interaction. Implemented a console UI and XML script engine for running tests. Also implemented a client/server-based UI for remote and distributed testing.
- Architect of a product-wide, cross-platform internationalization solution, based on dynamically-loaded XML string resources. This reduced translation work, since all resources were now in a common format, and also allowed Pervasive.SQL to utilize Japanese text on Linux in the supporting utilities for the first time in the history of the product.
- Implemented character codeset conversion routines for various European locales, Japanese Shift-JIS and Extended Unix Code (EUC-JP), and the various Unicode transformation formats (UTF8, UTF16, and UTF32).
- Authored an NLM import library generation utility, allowing the export of a binary C++ classes from Netware Loadable Modules. This removed the necessity of manually maintaining a text file containing the list of exports in C++ mangled names.
- Integrated Xerces XML SAX parser into the class library for seamless type matching, character conversion, and parser callback authoring.
- Developed a remote method invocation mechanism for our cross-platform COM implementation with protocol and transport pluggability. Provided scalable thread model, allowing natural transition from in-process to cross-process execution without behavioral changes.
- Other services implemented for class library include Blowfish encryption algorithm, MD5 secure hash algorithm, transport/channel network abstractions, and a compression library.
- Developed an SDK for Pervasive Services, including an installer, documentation, wizards, etc. Installer eased setup of developer build environment and configuration, including automatic tool detection and integration.
- Conducted performance analysis at both component and system levels, providing insight and solutions to solve performance problems, especially concerning SMP scalability.
- Acted as mentor for C++ and object oriented software design methodologies. Purveyor of best practices.

IBM - RS/6000 Server Development, Austin, Texas

System Engineer, June 1999 to March 2000

- Bring-up, debug, and test support of RS/6000 servers. Required skills include knowledge and analysis of computer system architecture, bus architectures, and hardware/firmware interactions.
- Responsible for first power-on of new systems, from service processor boot sequence, to main processor configuration and startup, and finally to main operating system hand-off. Debugged and fixed each level until next stage could be successfully reached.
- Component and interconnect debugging on mother boards, processor cards, and add-on cards.
- Located design and manufacturing defects using logic analyzers and oscilloscopes. Implemented fixes using available extraneous logic.

EDUCATION

Texas A&M University, College Station, Texas

Computer Engineering, B.S., May 1999, 3.896 GPR